Yaswanth Muppalla

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Senior Software Engineer | ***Java/Python Backend*** | ***Distributed Systems & APIs*** | ***AWS/GCP*** | ***AI Platforms***

PROFESSIONAL EXPERIENCE

Senior Software Engineer | American Express **Apr 2023 - Present**

Impact: Built a **cross-cloud data and API platform** processing **10M+ financial records**, improving reporting efficiency by **35%**, achieving **99% data accuracy**, and reducing rerun/debug cycles by **3x**.

- Engineered **JSON-configurable Java/Scala components** and **Kafka/Flink pipelines** for batch and streaming ingestion, joins, deduplication, validation, enrichment, and governed publishing; migrated the platform from **on-premises to GCP**.
- Automated **Airflow workflows** with dependency controls, **idempotent reruns**, **SLA monitoring**, data-quality gates, and **failure recovery** across high-volume reporting workloads.
- Published governed outputs to **BigQuery** and exposed analytical services through **AWS API Gateway, Lambda, and ECS** using **stable contracts, retries, access controls, and tracing**.
- Improved **platform reliability and auditability** through reconciliation checks, lineage, structured logging, failure diagnostics, and production support across **cross-cloud workflows**.
- Extended the data foundation with an **IRIS-backed RAG ingestion layer** for extraction, chunking, metadata enrichment, **scoped retrieval, citations, and grounded responses**.
- Built **MCP/API and agent-context services** for controlled tool execution, session state, prompt assembly, **guardrails, execution traces, and audit-ready source tracking**.

Software Engineer | Skechers **Oct 2020 - Mar 2023**

Impact: Delivered **15+ Python APIs and event-driven services**, maintained **99.9% uptime**, reduced infrastructure costs by **22%**, migrated **35M+ records**, and cut incident resolution time by **40%**.

- Developed **15+ Python/Pydantic REST APIs** on **AWS Lambda and API Gateway** for loyalty and campaign workflows, sustaining **99.9% uptime**.
- Orchestrated **event-driven processing** with **Step Functions**, retries, timeout handling, and downstream service calls, **isolating failures** across distributed loyalty operations.
- Applied **layered service architecture** across **validation, orchestration, data access**, retry handling, and downstream integrations.
- Migrated **35M+ SFMC records** to **MongoDB** and improved query performance by **50%** through indexing and caching.
- Integrated **SessionM, SFMC, MongoDB, and AWS services** for loyalty workflows with **sub-500ms transactions**.
- Improved **production resilience** with **CloudWatch/Datadog alerts**, **API failure tracing**, retries, and SLA-focused incident response.

Full-Stack Developer | SOCH Inc. - Richmond, VA **Jan 2020 - Sep 2020**

- Built a **Spring Boot ticketing platform** with **REST microservices**, validation, exception handling, and **service-layer boundaries** for government workflows.
- Designed **Spring REST services** with API controllers, **request validation, exception handling**, and **service-layer separation** for maintainable government workflows.
- Used **Java ExecutorService/CompletableFuture** for **asynchronous** ticket updates, notifications, and background job execution.
- Managed **concurrency** with **thread-safe collections**, **synchronization**, timeouts, and full-stack operational workflow screens.

FOUNDER-LED 0-TO-1 TECHNICAL PROJECT

OmInsights - Founder-Led Applied AI Platform Build **Jan 2024 - Present**

Impact: Built a **6-component AI platform** with **tenant-scoped APIs**, agent/context orchestration, MCP tool execution, async file processing, and **source-backed intelligence** with tenant isolation, citations, and traceability.

- Architected **Angular UI, FastAPI Business Service, Context Engine, MCP Server**, File Processing worker, and AWS deployment paths with CI/CD, environment promotion, secrets/config handling, and tenant-aware operations.
- Built an **AI evaluation layer** with groundedness, relevance, citation, and policy checks, **LLM-as-a-judge scoring**, and regression tracking.
- Designed **Context Engine flows** for **chat/session memory**, selected context, KB injection, prompt assembly, and **LLM/context harness design** for multi-turn agent orchestration, policy-aware grounding, and session continuity.
- Implemented **MCP/API runtime patterns** and a shared **LLM gateway across platform layers**, centralizing model/provider routing, request policies, retries/fallbacks, tool access, usage logging, and execution traces.
- Created **single-founder execution playbooks** across each platform layer for architecture decisions, implementation, deployment, testing, observability, and recovery, standardizing repeatable delivery and accelerating execution.

TECHNICAL SKILLS

Languages & Frameworks: Java 17, **Python**, Scala, SQL, **Spring Boot**, **FastAPI**, Pydantic, TypeScript/Angular
Backend & Distributed Systems: REST APIs, **microservices**, **event-driven architecture**, service contracts, **Java concurrency**, async workers, caching, retries, idempotency
Cloud & DevOps: AWS, **GCP**, Lambda, API Gateway, **ECS/Fargate**, Step Functions, S3/SQS, Docker, **AWS CDK/CloudFormation**, CI/CD
Data & Streaming: Kafka, **Flink**, **Airflow**, **BigQuery**, MemSQL, MongoDB, OpenSearch/Elasticsearch, batch/stream processing
Observability: Datadog, **CloudWatch**, Kibana, X-Ray, ELK Stack, **structured logging**, SLA alerts, failure tracing
AI Platform Systems: agent/context orchestration, **MCP/FastMCP**, **LLM gateways**, RAG/vector search, LLM-as-a-judge evaluations, guardrails, citations

EDUCATION

State University of New York, New Paltz | Master of Science, Computer Science | CGPA: 3.6/4.0 **May 2020**

JNTUK | Bachelor of Technology | CGPA: 8.5/10 **2018**